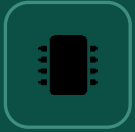




Vision-Verified Kitting & Sorting Robot

QR Mode Switching, WhatsApp Verification & Dobot Magician Execution

A robotic system that scans a QR code to select a task mode, verifies the required items are physically present using computer vision, sends WhatsApp notifications for anything missing, and autonomously executes the sorting task with the Dobot Magician once verification is complete.



Project Team

TEAM MEMBER

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PROJECT GUIDE

Prof. Suvidha Rupesh Kumar



Problem Statement

P-01

Manual verification of required parts/items before a task starts is slow and error-prone.

P-02

Incomplete kits or missing items often aren't noticed until a task has already failed or stalled.

P-03

No low-cost way for a robot to notify a user in real time about exactly what's missing.

P-04

Students lack exposure to integrating messaging/IoT APIs with real robotic task execution.

IMPACT

40%

efficiency loss from manual part verification

HARDWARE COST

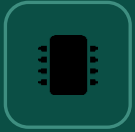
₹0

arm, conveyor & camera already in lab AB3-410

TASK MODES

2

QR-switchable kitting/sorting profiles



Motivation & Need

- 01 Reduce manual effort in verifying items before a task begins.
- 02 Give real-time visibility into what's missing, not after-the-fact failure.
- 03 Demonstrate real industrial automation with live task-mode switching.
- 04 Learn to integrate vision, messaging APIs, and robotic execution together.
- 05 Make full use of existing lab hardware for hands-on experience.

FASTER DETECTION

Real-time

missing-item alerts vs.
manual checks

VERIFY ACCURACY

~99%

achievable with CV + real
hardware

SETUP COST

₹0

hardware already in AB3–
410 lab



Objectives

OBJ-01

Detect a scanned QR code to select one of two task modes and load its required item checklist.

OBJ-02

Verify placed items in real time using YOLO/OpenCV against the required checklist.

OBJ-03

Notify the user via WhatsApp (Twilio API) of any items still missing until the kit is complete.

OBJ-04

Plan and execute the verified task autonomously with MoveIt + Dobot Magician once the checklist is complete.

OBJ-05

Support switching between two independent task modes by re-scanning a different QR code.

STACK**ROS 1 · MoveIt · YOLO · QR/OpenCV · Twilio WhatsApp**



SDG Alignment

UNITED NATIONS SDG

9

**Industry, Innovation
and Infrastructure**

*Build resilient infrastructure, promote inclusive
industrialisation and foster innovation.*

HOW THIS PROJECT ALIGNS

A-01

Supports automated kitting-verification patterns used in real manufacturing lines.

A-02

Demonstrates practical integration of IoT messaging with industrial robotics.

A-03

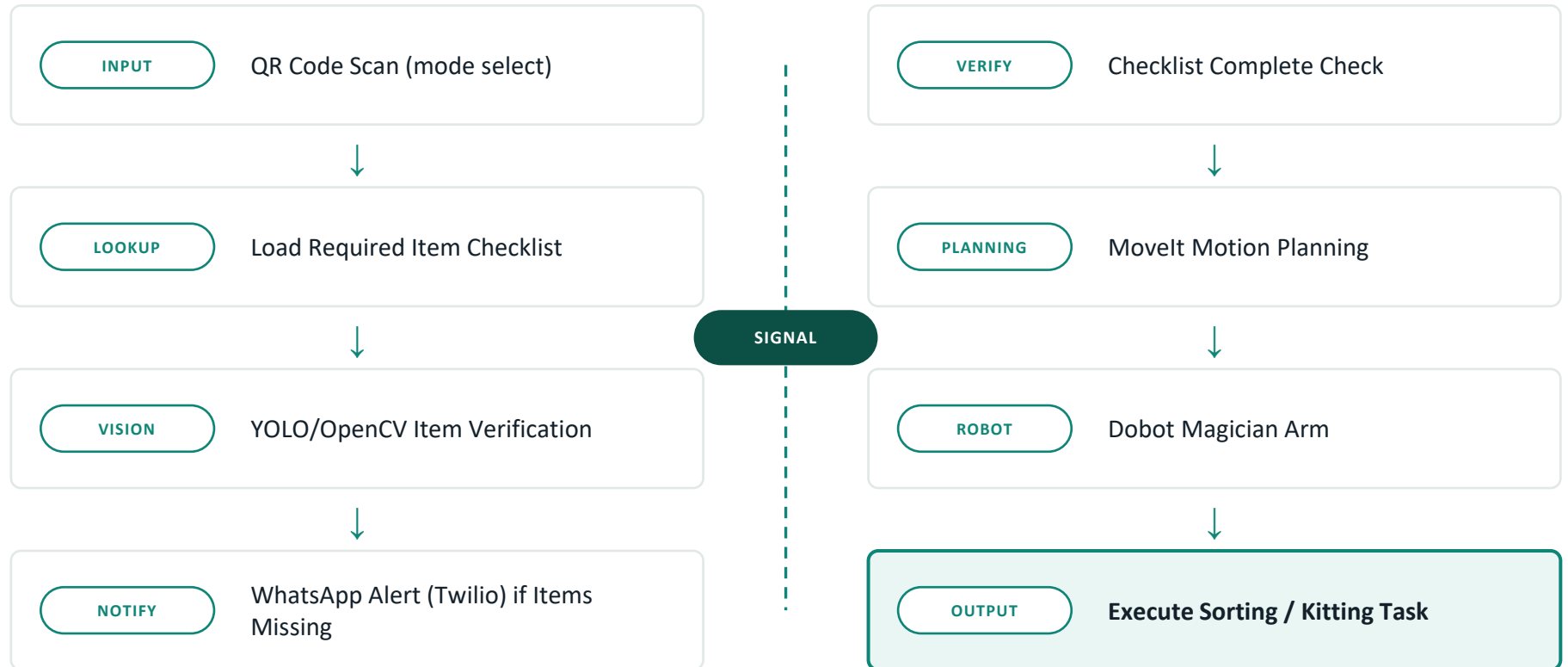
Promotes accessible robotics education using equipment already in the lab.

A-04

Reduces errors and waste from incomplete or misassembled kits before execution.



System Architecture





Feasibility Study

AX-01

Technical

- QR detection is built into OpenCV; Dobot ships an official ROS 1 driver (Magician_ROS).
- Hardware (arm, conveyor, vision kit) is already installed and tested in AB3-410.
- Twilio's WhatsApp Sandbox API provides free, ready-to-use messaging integration.

✓ FEASIBLE

AX-02

Economic

- Zero additional hardware cost — arm, conveyor, and camera already provided by the lab.
- All software (ROS 1, MoveIt, OpenCV, YOLO, Twilio sandbox) is free.
- Only minor consumables needed: kit items, QR printouts, and small bins.

✓ FEASIBLE

AX-03

Operational

- Tabletop hardware setup — safe to run with no injury risk.
- Requires lab wifi/internet access for WhatsApp notifications to send.
- Live demo is easy to follow: scan, place items, get notified, watch it execute.

✓ FEASIBLE



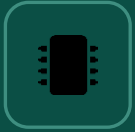
System Requirements

HARDWARE — AVAILABLE IN LAB AB3-410

ARM	Dobot Magician Educational Kit 4-DOF manipulator for pick-and-place
VISION	Vision Kit Camera Captures live feed for QR scan + item detection
CONVEYOR	Mini Conveyor Belt Kit Delivers items into arm's reach
COMPUTE	Jetson Orin Nano / Dell Workstation Runs YOLO, QR detection & ROS nodes
NETWORK	Lab Wi-Fi / Internet Required for WhatsApp (Twilio) messaging

SOFTWARE — TO INSTALL

OS	Ubuntu 20.04 LTS Required for ROS 1 Noetic
MIDDLEWARE	ROS 1 (Noetic) Core communication framework
DRIVER	Magician_ROS Official Dobot ROS 1 driver
PLANNING	Movelt (1) Motion planning framework
VISION	YOLO + OpenCV (QR detection) Item detection & mode scanning
MESSAGING	Twilio WhatsApp API (Python SDK) Sends missing-item notifications



Estimated Cost

ITEM	COST	NOTE
Dobot Magician + Conveyor + Vision Kit	Existing (Lab)	Already installed in AB3–410
Compute (Jetson / Workstation)	Existing (Lab)	No new purchase needed
Ubuntu 20.04 + ROS 1 Noetic	Free	Open-source
MoveIt (1) + OpenCV (QR + item detection)	Free	Open-source, BSD / Apache 2.0
Twilio WhatsApp API	Free	Sandbox tier, sufficient for demo use
Kit items, QR printouts & bins	~₹200–500	Sorting objects, printed QR cards, small bins
TOTAL ADDITIONAL COST	~ ₹200–500	excl. lab-provided hardware

SOFTWARE COST

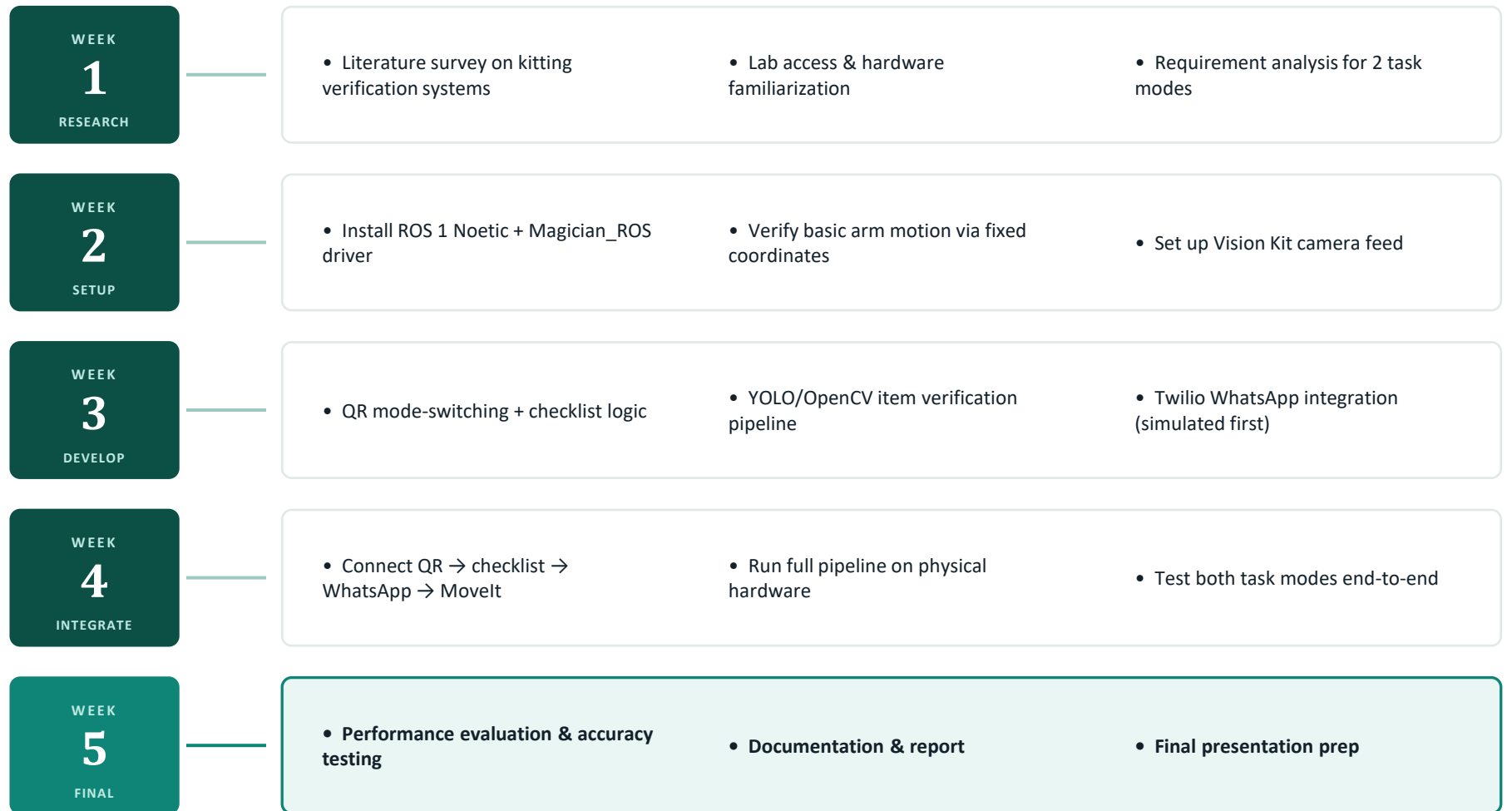
₹0

100% open-source stack

LICENSES

- ✓ Apache 2.0 (ROS 1)
- ✓ BSD (MoveIt)
- ✓ AGPL-3.0 (YOLO)
- ✓ Free Sandbox (Twilio API)

Project Timeline





Expected Outcomes

OUT-01

Reliable QR-based switching between two independent task modes.

OUT-02

Accurate real-time verification of placed items against a required checklist.

OUT-03

Automated WhatsApp notifications for missing items until the kit is complete.

OUT-04

Autonomous task execution via MoveIt + Dobot Magician only after verification.

OUT-05

End-to-end demonstration of vision + messaging + robotic execution pipeline.

OUT-06

Reusable foundation for extending to more task modes / industrial kitting checks.

STACK ROS 1 · MoveIt · YOLO · OpenCV · Twilio WhatsApp · Dobot Magician

PROJECT PLAN COMPLETE ✓